1. Given $f(x) = -x^2 + 3x - 9$, find $f(-6)$

$\boxed{-63}$

2. Express as a trinomial: $(x+9)(x-2)$

$\boxed{x^2 + 7x - 18}$

3. Factor completely: $49 - 16x^2$

$\boxed{(7+4x)(7-4x)}$

4. Factor the trinomial: $5x^2 + 16x + 12$

$\boxed{(5x+6)(x+2)}$

5. Express the following fraction in simplest form using only positive exponents.

$$\frac{5(y^5)^3}{2y^{10}}$$

$$\boxed{\frac{5y^5}{2}}$$

6. Express in simplest radical form: $\sqrt{28}$

$\boxed{2\sqrt{7}}$

7. The sum of three consecutive odd integers is 27. Find the value of the least of the three.

$\boxed{7}$

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8. Solve for x:

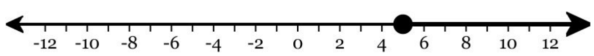
$$-11x - 7 = -10x - 17$$

$$\boxed{x = 10}$$

9. Solve the inequality and graph the solution on the line provided.

$$-7x + 2 \leq -33$$

$$x \geq 5$$



10. Solve the following equation for D.

$$Dm^3 + n = r$$

$$\boxed{D = \frac{\mathbf{r - n}}{\mathbf{m^3}}}$$

11. Solve the following equation for x. Express your answer in the simplest form. If there are infinite solutions state "infinite solutions" and if there are no solutions state "no solutions."

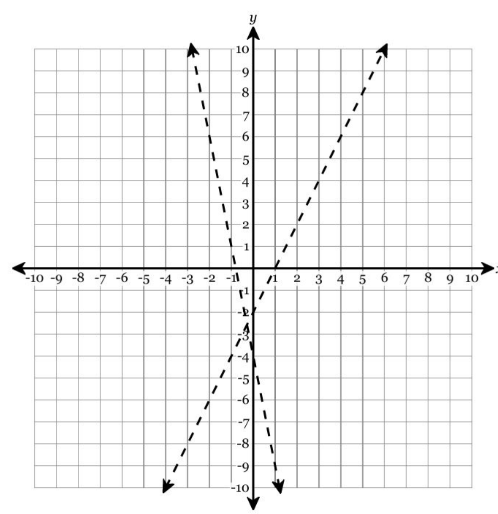
$$-8x + 8 = -8x + 8$$

The equation has infinitely many solutions.

12. Below are two inequalities and the graphs of their lines *without* the shading. By imagining where the shading should be, identify which point would satisfy BOTH inequalities.

$$y > -5x - 4$$

$$y > 2x - 2$$



A. $(-10, 9)$

B. $(-1, -6)$

C. $(2, -1)$

D. $(2, 7)$

13. What is the discriminant of the quadratic equation $5x^2 - 8x + 5 = 0$?

$\boxed{\text{A. } -36}$ B. 36 C. 164 D. -164

14. What is the slope of the line that passes through the points $(-9, 5)$ and $(-9, 0)$? Write your answer in *simplest form*.

$$\boxed{\text{Undefined}}$$

15. Put the following equation of a line into slope-intercept form, simplifying all fractions.

$$20x + 24y = 192$$

$$\boxed{y = -\frac{5}{6}x + 8}$$

16. What is the equation of the line that passes through the point $(3, -8)$ and has a slope of 0?

$$\boxed{y = -8}$$

17. Find the equation of a line parallel to $-6x + y = -6$ that passes through the point $(1, -2)$.

A. $-6x - y = -4$

$\boxed{\text{B. } y = 6x - 8}$

C. $y = -6x - 8$

D. $-x - 6y = 11$

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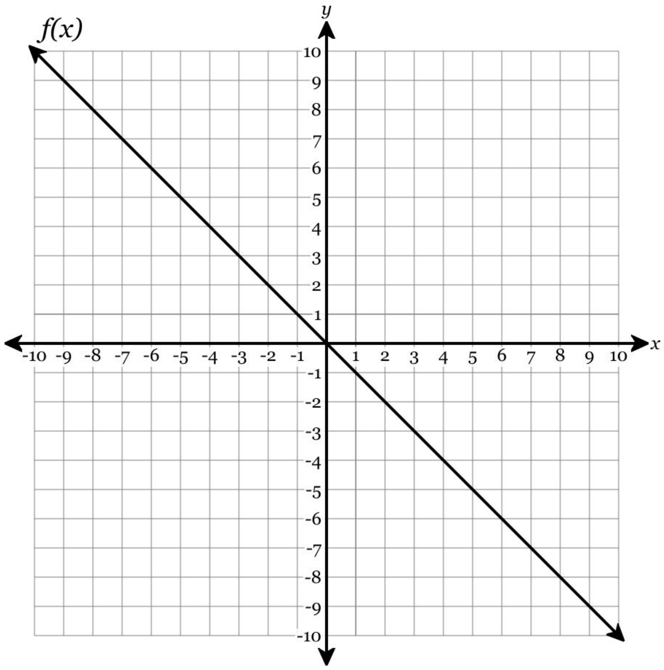
18. Find $h(-7)$.



$$g(x) = x^2 - 4x - 5$$

x	$h(x)$
-7	-8
7	2
8	-1
-10	8
-3	3
5	10

$\boxed{-8}$

19. Which set of ordered pairs represents a function?

A. $\{(9,3),(0,3),(8,9),(-5,5)\}$

B. $\{(0,-6),(1,8),(-7,-7),(-7,6)\}$

C. $\{(1,-9),(-3,9),(-4,-5),(-3,3)\}$

D. $\{(9,-9),(4,9),(4,3),(-1,1)\}$

20. The number of newly reported crime cases in a county in New York State is shown in the accompanying table, where x represents the number of years since 2001, and y represents number of new cases. (a) Write the linear regression equation that represents this set of data, rounding all coefficients *to the nearest hundredth*.
(b) Using this equation, estimate the *calendar* year in which the number of new cases would reach 467.

x	0	1	2	3	4	5
y	788	740	705	673	613	633

$y = -33.94x + 776.86$

2010

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